

Georgia Zhang

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EDUCATION

University of Toronto

Toronto, ON

BASc., Computer and Electrical Engineering + **PEY Co-op**; GPA: 3.73/4.0, Dean's List Sep. 2024 – May 2029

- Relevant Coursework: Digital Systems, Computer Organization, Hardware Design and Communication, Digital Electronics, Computer Architecture, Operating Systems

EXPERIENCE

Embedded Systems Student Intern

May 2026 – Aug. 2026

Network Research Lab, University of Toronto

Toronto, ON

- Used **FreeRTOS** to implement concurrent sensor collection and cloud transmission tasks, coordinating with queues and sleep states to reduce power use by 20%.
- Designed and built a **4-layer electrochemical breakout board** using the **LMP91000** and **ADS122C04 24-bit ADC** to measure **nanoamp currents** from electrochemical sensors, achieving 0.85 pA resolution.
- Resolved a **GPIO12 strapping pin** conflict by delaying software serial initialization 10s, eliminating recurring ESP32 boot failures.
- Eliminated ESP32 brownouts during modem TX by powering the modem from battery instead of the **LDO**, resolving current-limit failures.

Firmware Developer

May 2026 – Present

University of Toronto Aerospace Team (UTAT) – Space Systems

Toronto, ON

- Configured **MCUboot** on **Zephyr RTOS** for **STM32G431RB**, fixing DTS/flash errors to enable reliable firmware updates.
- Used **SSH** and the **UTAT VPN** to remotely flash STM32 firmware through a Raspberry Pi using PyOCD.

VP of Partnerships

May 2026 – Present

University of Toronto Robotics Association (UTRA)

Toronto, ON

- Created **sponsorship packages** and technical reports to secure monetary and in-kind industry support.
- Coordinated parts, components, and funding across **Combat, ART, PacBots, RoboSoccer, and SUMO**.

SUMO Bot Team Project Manager

May 2025 – May 2026

University of Toronto Robotics Association (UTRA)

Toronto, ON

- Led firmware and hardware setup sessions for **30+ teams** of 3 engineers building autonomous **SUMO robots**.
- Ran **Arduino C++** and hardware workshops covering sensors, wiring, and soldering to onboard new members.
- Coordinated a SUMO tournament with **York, Ontario Tech, TMU, and McMaster** to manage event logistics.

PROJECTS

Digital Logic Simulator & Compiler

Mar. 2026 – Apr. 2026

- Built a bare-metal **C** digital logic simulator on **RISC-V** to execute combinational and sequential circuits on the **DE1-SoC** in real time.
- Implemented **VGA** graphics and 90-degree wire routing to visualize circuit structure and live logic states.
- Developed a **C#** compiler to translate text-based circuit definitions into binary instructions, enabling custom circuits to run on **FPGA memory**.
- Integrated **PS/2** keyboard input to modify circuit inputs during execution and generate truth tables.

Vehicle Lighting Control Board

Dec. 2025 – Mar. 2026

- Designed a **KiCad** PCB with 12V, 5V, and 3.3V rails to power an **ESP32** for vehicle lighting control.
- Implemented **C++** firmware to read IMU orientation, triggering lights when rollover thresholds were exceeded.
- Added **CAN bus** communication to exchange vehicle state with other control boards, enabling coordinated lighting response.

TECHNICAL SKILLS

Languages: C, C++, C#, Assembly, Verilog

Embedded Systems & RTOS: STM32, ESP32, RISC-V, FreeRTOS, Zephyr RTOS, MCUboot, CAN, I2C, UART

Hardware & PCB: KiCad, Multilayer PCB Design

Software & Tools: Git, Linux, SSH, PyOCD, AWS IoT, InfluxDB, Grafana